

Oct.-16/T.E./Insem.-41
T.E. (Computer) (Semester - I)
OPERATING SYSTEM DESIGNS
(2012 Pattern)

*Time : 1 Hour]**[Max. Marks : 30**Instructions to the candidates:*

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

- Q1)** a) Explain with neat diagram the importance of buffer cache. [5]
 b) Explain getblock () algorithm. [5]

OR

- Q2)** a) Explain structure of regular files in UNIX System V. [4]
 b) Explain namei algorithm. [4]
 c) Write a short note on Super Block. [2]

- Q3)** a) Draw and Explain Process State transition Diagram. [6]
 b) Explain context of a process. [4]

OR

- Q4)** a) Solve the following by Banker's Algorithm [6]

	Allocation Matrix	Max Claim Matrix	Total Resources
	A B C D	A B C D	A B C D
P0	0 1 1 0	0 2 1 0	3 17 16 12
P1	1 2 3 1	1 6 5 2	
P2	1 3 6 5	2 3 6 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

- Create the need matrix.
 - Use the safety algorithm to test if the system is in a safe state.
- b) Explain fork () system call. [4]

P.T.O.

- Q5)** a) What is swapping? How allocation and freeing of swap space is done. [5]
b) Explain buddy system with example. [5]

OR

- Q6)** a) Explain Translation Lookaside Buffer. [5]
b) Explain in brief what is paging and segmentation? How Logical-to-Physical Address Translation is done in both? [5]

